# Akoizo Original Practice - Form 5

### Division B | Scrambler | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

**Name: _____ Date: _____**

**1. (2 points) Which egg holder detail is NOT legal?**

Unit: Protecting the egg.
Worked-case data: A vehicle arrives at 1.5 m/s and stops in 6 cm. Find the average deceleration in g.
Separate investigation observations:
Design A: Arrives 3.0 m/s, stops in 2 cm; 2 of 5 eggs cracked.
Design B: Arrives 1.5 m/s, stops in 5 cm; 0 of 5 cracked.
Question: Compare decelerations.

A. A balsa backstop

B. A 3/8 inch dowel

C. Dowels 3.5 cm long

D. Dowel bottoms 7 cm above the track

Answer: _____

**2. (2 points) What happens if string piles up on the axle?**

Unit: Drivetrains: turning a drop into distance.
Worked-case data: A vehicle has 0.50 cm axle radius and 4.0 cm wheel radius. The mass pulls 0.75 m of string. About how far can the vehicle be driven?
Separate investigation observations:
Run 1: String wound in one neat layer: 8.42 m.
Run 2: String piled in three layers: 7.61 m.
Same: Mass and drop height.

A. Effective axle radius grows, changing distance per centimeter of string

B. Friction disappears

C. The wheel shrinks

D. Nothing

Answer: _____

**3. (2 points) Which is the only allowed source of propulsion energy?**

Unit: Rules and the low-score system.
Worked-case data: A run stops 12.4 cm from the Target Point in 5.31 s with an Inside Can Distance of 60.0 cm and no violations. Find the Run Score.
Separate investigation observations:
Vehicle A: Stops 20 cm away in 3.0 s.
Vehicle B: Stops 6 cm away in 7.5 s.
Both: No cans, no violations.

A. A falling mass up to 2.00 kg

B. A wound rubber band

C. A battery motor

D. A compressed spring

Answer: _____

**4. (2 points) When is the Can Bonus lost?**

Unit: The Can Bonus and curved paths.
Worked-case data: Target 9.00 m. The team sets a 60 cm gap. Find the bonus, the gap center offset, and the arc radius.
Separate investigation observations:
Arc runs: Passed between cans 10 of 10.
Stopping: Stopped 18 cm short of the target each time.
Setting: Turns calculated from the straight-line distance.

A. If the egg is foremost

B. If the gap is over 50 cm

C. If any part of the vehicle does not pass between the cans or a can is moved

D. If the run takes over 5 s

Answer: _____

**5. (4 points) Explain how a wing nut mechanism sets distance.**

Unit: Automatic stopping.
Worked-case data: Target 9.25 m. Wheels are 7.50 cm in diameter. The vehicle skids 3 cm after braking, and the egg tip sits 10 cm ahead of the point used as the start reference. How many turns?
Separate investigation observations:
Setting 30.0 turns: Stops at 7.21, 7.24, 7.19 m.
Setting 35.0 turns: Stops at 8.40, 8.43, 8.38 m.
Target: 8.00 m.

Answer: _____
_____
_____

### 6. (4 points) Why is consistent release important?

Unit: Energy from a falling mass.
Worked-case data: A 1.50 kg mass falls 0.60 m. If 40 percent reaches a 0.50 kg vehicle as kinetic energy, what is its top speed?
Separate investigation observations:
2.0 kg mass: Top speed 4.2 m/s; egg cracked on stop in 3 of 5 runs.
1.2 kg mass: Top speed 2.6 m/s; egg safe 5 of 5; time +1.8 s.
Target: 8.00 m in both cases.

Answer: _____
_____
_____

### 7. (4 points) Why may participants only touch the pencil during triggering?

Unit: Calibration and competition day.
Worked-case data: Table: 30 turns -> 7.20 m, 34 turns -> 8.15 m, 38 turns -> 9.10 m. Set for a State target of 8.60 m.
Separate investigation observations:
Run 1: Straight run: 4.2 cm from target, 6.1 s, no cans.
Run 1 score: 100 + 8.4 + 6.1 = 114.5.
Option for Run 2: 60 cm can gap; bonus runs average 9 cm error, 6.5 s.

Answer: _____
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_____

### 8. (4 points) What violations involve the Terminal Barrier?

Unit: Protecting the egg.
Worked-case data: A vehicle arrives at 1.5 m/s and stops in 6 cm. Find the average deceleration in g.
Separate investigation observations:
Design A: Arrives 3.0 m/s, stops in 2 cm; 2 of 5 eggs cracked.
Design B: Arrives 1.5 m/s, stops in 5 cm; 0 of 5 cracked.
Question: Compare decelerations.

Answer: _____
_____
_____

### 9. (4 points) Why release the string when fully unwound?

Unit: Drivetrains: turning a drop into distance.
Worked-case data: A vehicle has 0.50 cm axle radius and 4.0 cm wheel radius. The mass pulls 0.75 m of string. About how far can the vehicle be driven?
Separate investigation observations:
Run 1: String wound in one neat layer: 8.42 m.
Run 2: String piled in three layers: 7.61 m.
Same: Mass and drop height.

Answer: _____
_____
_____

**10. (6 points) Solve this case and explain your reasoning: A run stops 12.4 cm from the Target Point in 5.31 s with an Inside Can Distance of 60.0 cm and no violations. Find the Run Score.**

Answer: _____
_____
_____

**11. (6 points) Solve this case and explain your reasoning: Target 9.00 m. The team sets a 60 cm gap. Find the bonus, the gap center offset, and the arc radius.**

Answer: _____
_____
_____

**12. (2 points) About what setting should the team use?**

Supplied synthetic observations:
Setting 30.0 turns: Stops at 7.21, 7.24, 7.19 m.
Setting 35.0 turns: Stops at 8.40, 8.43, 8.38 m.
Target: 8.00 m.

A. About 33.3 turns

B. 32.0 turns

C. 35.0 turns

Answer: _____

**13. (2 points) Define “Gravitational potential energy” in the context of Scrambler.**

Answer: _____
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_____

**14. (2 points) Define “Target Distance” in the context of Scrambler.**

Answer: _____
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**15. (2 points) Define “Dowels” in the context of Scrambler.**

Answer: _____
_____
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**16. (2 points) Define “Wheel circumference” in the context of Scrambler.**

Answer: _____
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**17. (2 points) Define “Distance Score” in the context of Scrambler.**

Answer: _____
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**18. (2 points) Define “Inside Can Distance” in the context of Scrambler.**

Answer: _____
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